FITTING A GEV IN JULIA

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CEVE 543

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Port Pirie data

Maximum likelihood by hand

L-moment estimation

Gumbel vs GEV

Diagnostics

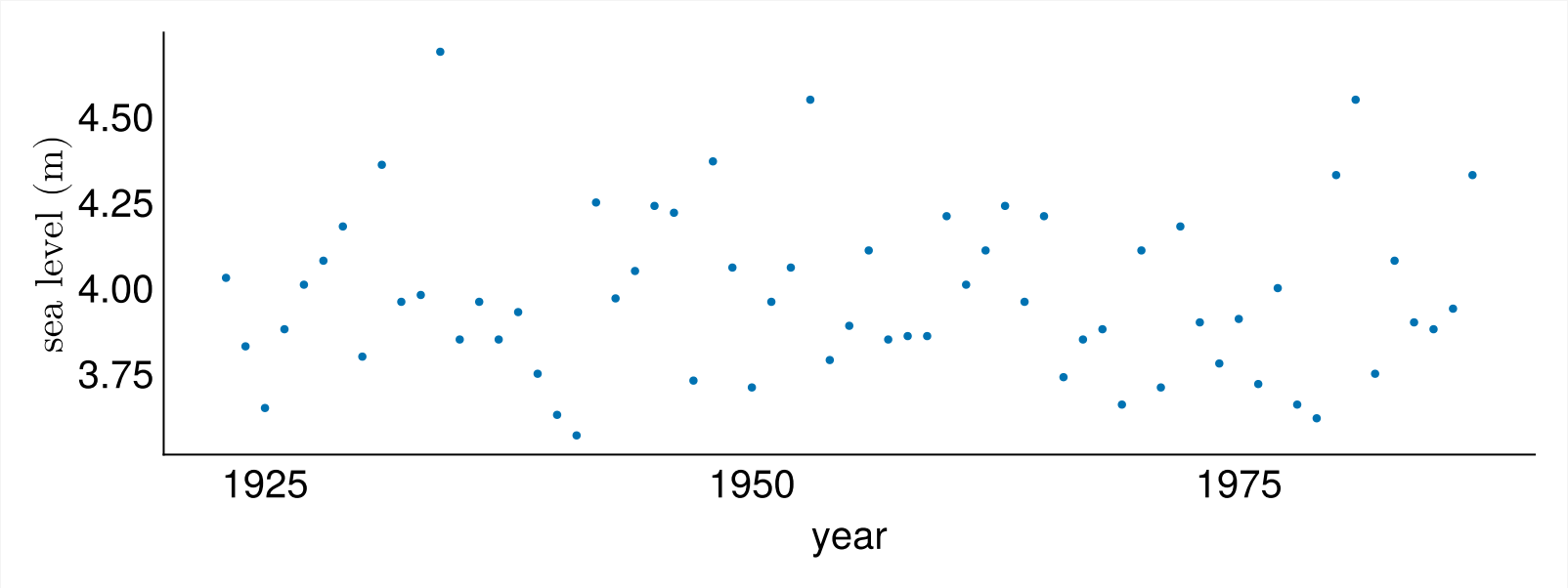
Bayesian inference

ANNUAL MAXIMUM SEA LEVELS AT PORT PIRIE

65 years of annual maximum sea level (metres) at Port Pirie, South Australia, 1923–1987 (Coles, 2001, §3.4).

```
portpirie = CSV.read(joinpath(here, "portpirie.csv"), DataFrame)
y = portpirie.SeaLevel
fig = Figure(; size=(800, 300))
ax = Axis(fig[1, 1]; xlabel="year", ylabel=L"\text{sea level (m)}")
scatter!(ax, portpirie.Year, y; color=colors[1], markersize=6)
fig
```

ANNUAL MAXIMUM SEA LEVELS AT PORT PIRIE

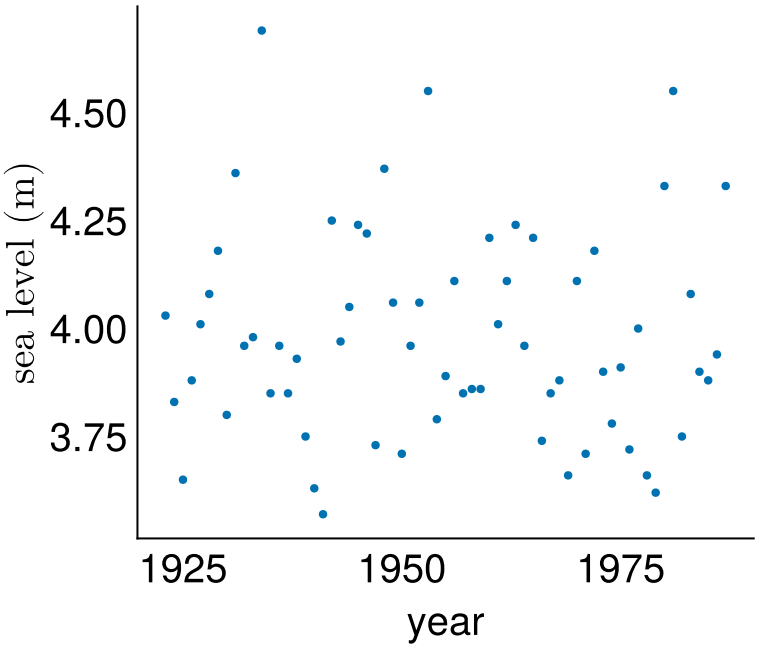


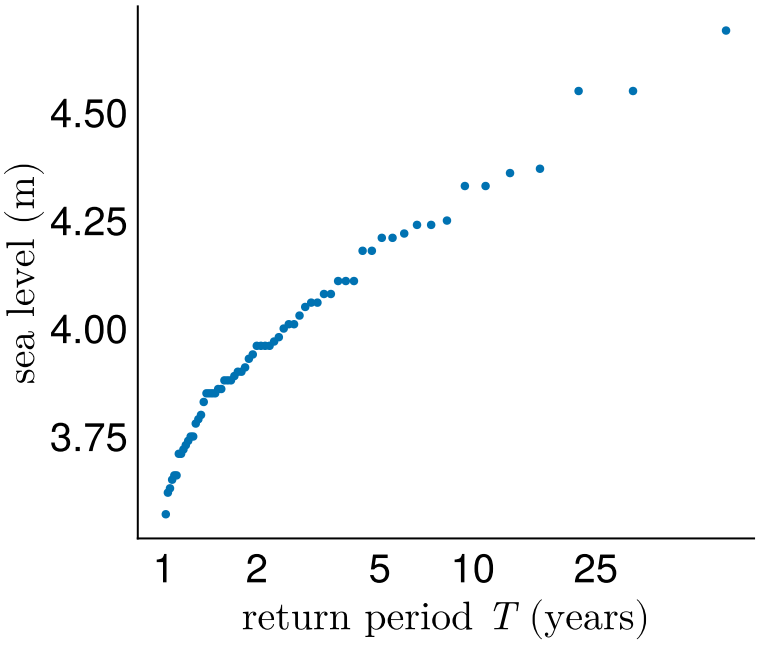
THE DATA ON A RETURN-PERIOD SCALE

Rank the data and assign Weibull plotting positions $\hat{T}_i = (n+1)/i$.

```
sorted, p_hat, T_hat = plotting_positions(y)
fig = Figure(; size=(800, 350))
ax1 = Axis(fig[1, 1]; xlabel="year", ylabel=L"\text{sea level (m)}")
scatter!(ax1, portpirie.Year, y; color=colors[1], markersize=6)
ax2 = Axis(fig[1, 2]; ylabel=L"\text{sea level (m)}")
linkyaxes!(ax1, ax2)
scatter!(ax2, T_hat, sorted; color=colors[1], markersize=6)
return_period_axis!(ax2)
fig
```

THE DATA ON A RETURN-PERIOD SCALE





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THE GEV LOG-DENSITY

From the board: the GEV has three parameters (μ, σ, ξ) and its log-density is

$$\log f(z \mid \mu, \sigma, \xi) = -\log \sigma - (1 + 1/\xi) \log t - t^{-1/\xi}$$

where $t = 1 + \xi(z - \mu)/\sigma > 0$.

THE NEGATIVE LOG-LIKELIHOOD

MLE finds the parameter vector that maximises the sum of log-densities, or equivalently minimises the negative.

```julia
function gev_nll(θ, y)
    μ, logσ, ξ = θ
    σ = exp(logσ)                       # optimise in log σ so scale stays positive
    d = GeneralizedExtremeValue(μ, σ, ξ)
    nll = 0.0
    for yᵢ in y
        lp = logpdf(d, yᵢ)
        isfinite(lp) || return 1e10     # penalty for out-of-support proposals
        nll -= lp
    end
    return nll
end
```

OPTIMISING WITH NELDER-MEAD

```
θ₀ = [mean(y), log(std(y)), 0.1]      # initial guess: sample moments, slight positive shape
result = optimize(θ -> gev_nll(θ, y), θ₀, NelderMead())
μ̂, logσ̂, ξ̂ = Optim.minimizer(result)
σ̂ = exp(logσ̂)
@printf("μ̂ = %.3f, σ̂ = %.3f, ξ̂ = %.3f", μ̂, σ̂, ξ̂)
```

```
μ̂ = 3.875, σ̂ = 0.198, ξ̂ = -0.050
```

THE CONVENIENCE WRAPPER

`CEVE543Utils.gevfit` wraps the same optimization with support for covariates.

```
1 fit_gev = gevfit(y)
2 d_gev = only(getdistribution(fit_gev))
3 @printf("gevfit: μ̂ = %.3f, σ̂ = %.3f, ξ̂ = %.3f", d_gev.μ, d_gev.σ, d_gev.ξ)
```

```
gevfit: μ̂ = 3.875, σ̂ = 0.198, ξ̂ = -0.050
```

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L-MOMENTS

MLE requires an optimizer and can be sensitive to starting values or small samples. L-moments are linear combinations of order statistics that give closed-form parameter estimates: no optimizer, always finite, and often more stable in small samples.

They are estimated from probability weighted moments $b_r = \frac{1}{n}\sum_{i=1}^{n} y_{(i)}\binom{i-1}{r}/\binom{n-1}{r}$.

The first three L-moments:

$$\ell_1 = b_0 \qquad \ell_2 = 2b_1 - b_0 \qquad \ell_3 = 6b_2 - 6b_1 + b_0$$

ℓ_1 is the mean, ℓ_2 measures spread, and the L-skewness ratio $\tau_3 = \ell_3/\ell_2$ measures asymmetry.

FROM L-MOMENTS TO GEV PARAMETERS

The shape follows from the PWM ratios through the polynomial approximation of Hosking, Wallis & Wood (1985):

$$c = \frac{2b_1 - b_0}{3b_2 - b_0} - \frac{\ln 2}{\ln 3} \qquad k = 7.859c + 2.955c^2 \qquad \xi = -k$$

Then scale and location follow in closed form:

$$\sigma = \frac{k(2b_1 - b_0)}{(1 - 2^{-k})\,\Gamma(1 + k)} \qquad \mu = b_0 - \frac{\sigma}{k}(1 - \Gamma(1 + k))$$

```julia
d_lmom = gevfit_lmom(y)
@printf("L-moments: μ̂ = %.3f, σ̂ = %.3f, ξ̂ = %.3f", d_lmom.μ, d_lmom.σ, d_lmom.ξ)
```

FROM L-MOMENTS TO GEV PARAMETERS

L-moments: $\hat{\mu} = 3.873$, $\hat{\sigma} = 0.203$, $\hat{\xi} = -0.051$

TWO ESTIMATORS, TWO ANSWERS

```
MLE:       μ = 3.875, σ = 0.198, ξ = -0.0501
L-moment: μ = 3.873, σ = 0.203, ξ = -0.0515
```

Different estimators give different answers on the same data, especially for ξ. The shape parameter is the hardest to estimate and the most consequential for the tail.

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DOES THE SHAPE PARAMETER MATTER?

The Gumbel is the GEV with $\xi = 0$ forced: the tail is exponential by assumption.

```julia
T_grid = exp.(range(log(1.1), log(500); length=100))
rl_gev = [quantile(d_gev, 1 - 1 / T) for T in T_grid]

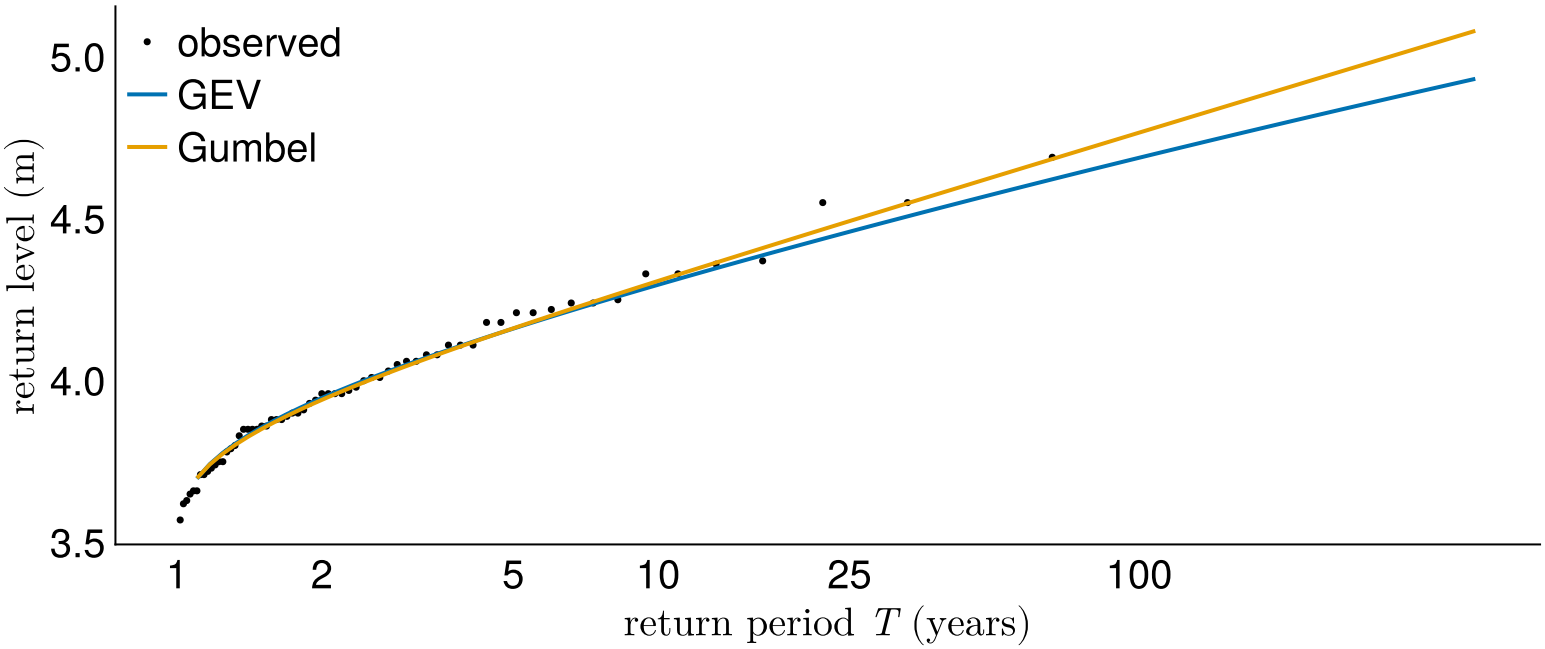
function gumbel_nll(θ, y)
    μ, logσ = θ
    σ = exp(logσ)
    d = Gumbel(μ, σ)
    nll = 0.0
    for yᵢ in y
        lp = logpdf(d, yᵢ)
        isfinite(lp) || return 1e10
        nll -= lp
```

DOES THE SHAPE PARAMETER MATTER?

```
    end
    return nll
end
res_gumbel = optimize(θ -> gumbel_nll(θ, y), [mean(y), log(std(y))], NelderMead())
d_gumbel = Gumbel(Optim.minimizer(res_gumbel)[1], exp(Optim.minimizer(res_gumbel)[2]))
rl_gumbel = [quantile(d_gumbel, 1 - 1 / T) for T in T_grid]
@printf("Gumbel MLE: μ̂ = %.3f, σ̂ = %.3f", d_gumbel.μ, d_gumbel.θ)
```

```
Gumbel MLE: μ̂ = 3.869, σ̂ = 0.195
```

THE TWO CURVES DIVERGE IN THE TAIL



They agree at short return periods and diverge past 50 years.

WHICH FIT SHOULD YOU USE?

$\hat{\xi} \approx -0.05$: the GEV bends slightly downward, the Gumbel stays straight.

Coles (Coles, 2001, §3.5.1): the Gumbel's confidence interval on the 100-year level is narrower because forcing $\xi = 0$ removes a source of uncertainty. The data do not rule out $\xi \neq 0$, so the GEV's wider interval is the one that reflects what the data actually say.

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THE QQ PLOT

Sort the observed data from smallest to largest: $y_{(1)} \leq y_{(2)} \leq \cdots \leq y_{(n)}$. These are the **observed quantiles**.

The **model quantiles** come from the fitted CDF. The i-th model quantile is the value the fitted distribution predicts the i-th smallest observation should take:

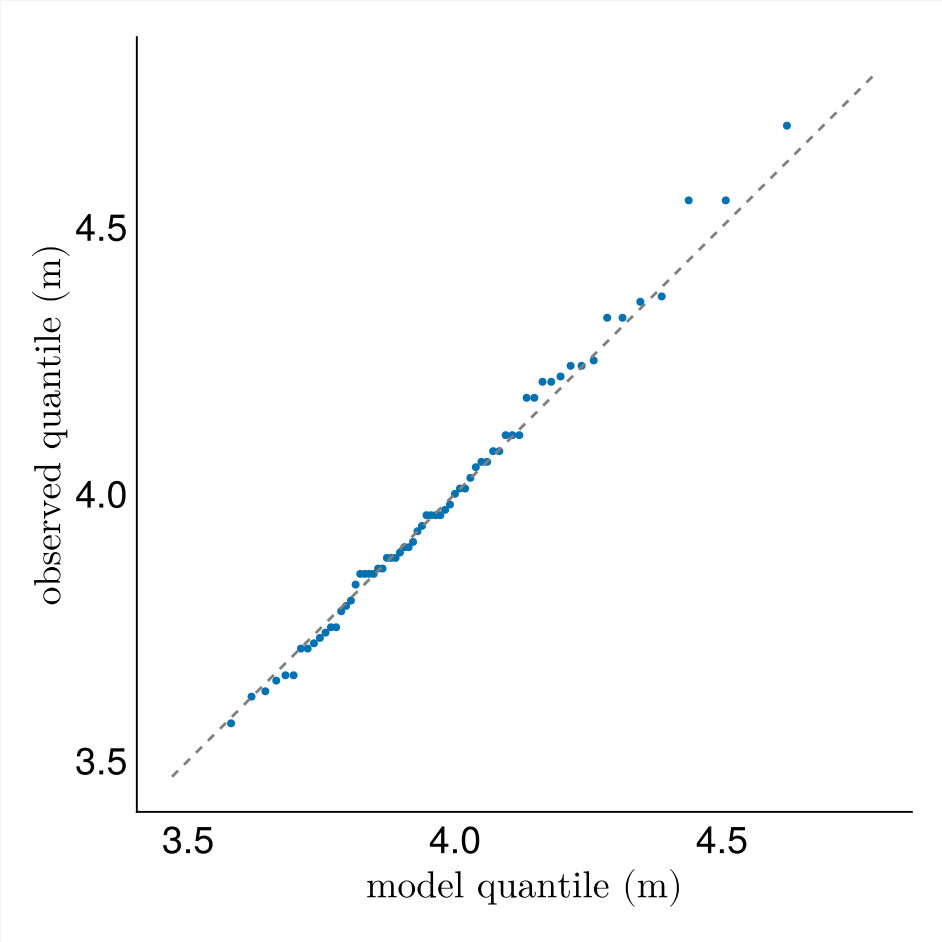
$$q_i = F^{-1}\left(\frac{i}{n+1}; \hat{\mu}, \hat{\sigma}, \hat{\xi}\right)$$

Plot one against the other. If the model fits well, each observed quantile is close to its model quantile, and the points fall near the diagonal.

QQ PLOT FOR PORT PIRIE

```
n_obs = length(y)
y_sorted = sort(y)
model_q = [quantile(d_gev, i / (n_obs + 1)) for i in 1:n_obs]
fig = Figure(; size=(500, 500))
ax = Axis(fig[1, 1]; xlabel=L"\text{model quantile (m)}",
    ylabel=L"\text{observed quantile (m)}", aspect=1)
scatter!(ax, model_q, y_sorted; color=colors[1], markersize=6)
lims = (minimum(y_sorted) - 0.1, maximum(y_sorted) + 0.1)
lines!(ax, collect(lims), collect(lims); color=:gray, linestyle=:dash)
fig
```

QQ PLOT FOR PORT PIRIE

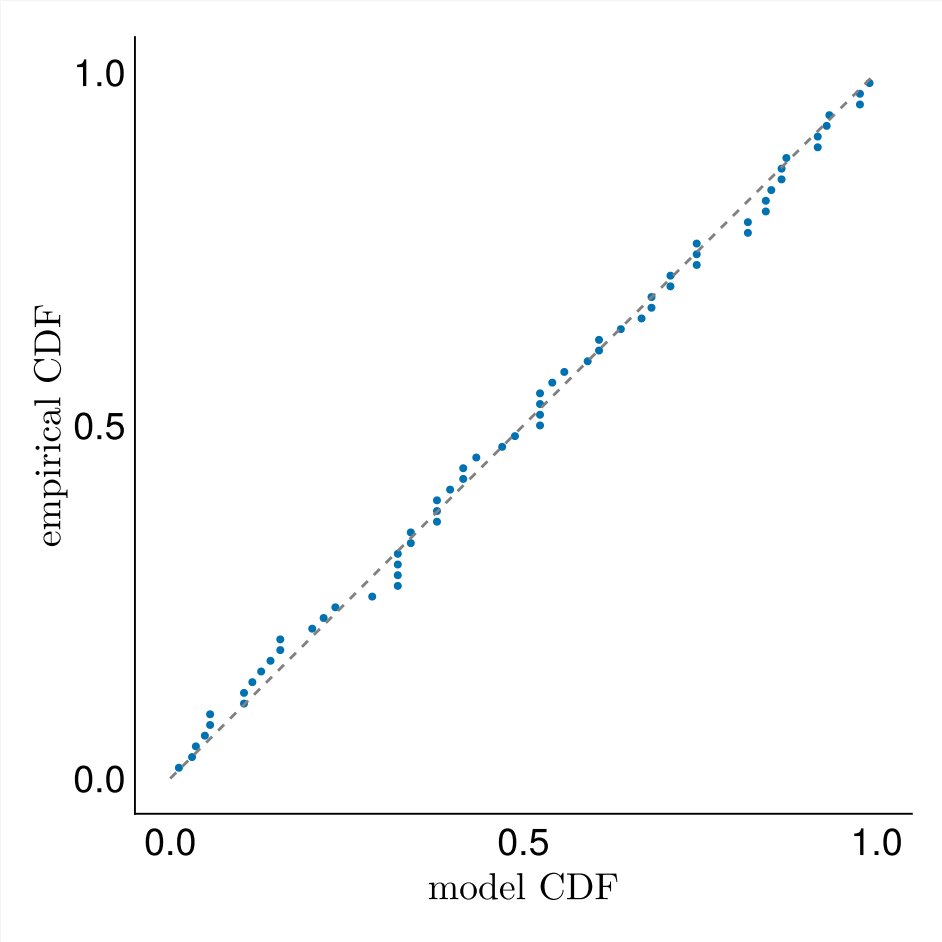


THE PROBABILITY PLOT

Both axes on the CDF scale: $\hat{F}$ vs $F_{\hat{\theta}}$. The probability plot checks the body of the distribution, where most of the data sit. It compresses the tail, so use the QQ plot for the upper end.

```
F_empirical = (1:n_obs) ./ (n_obs + 1)
F_model = [cdf(d_gev, y_sorted[i]) for i in 1:n_obs]
fig = Figure(; size=(500, 500))
ax = Axis(fig[1, 1]; xlabel=L"\text{model CDF}",
    ylabel=L"\text{empirical CDF}", aspect=1)
scatter!(ax, F_model, F_empirical; color=colors[1], markersize=6)
lines!(ax, [0, 1], [0, 1]; color=:gray, linestyle=:dash)
fig
```

THE PROBABILITY PLOT



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WHY BAYES?

MLE gives one $\hat{\theta}$ and one return-level curve. With 65 years of data, the parameters carry real uncertainty: a different 65-year draw from the same process would give a different $\hat{\theta}$. A Bayesian posterior is a distribution over θ that reflects that uncertainty.

Bayes' theorem:

$$p(\theta \mid y) \propto p(y \mid \theta)\, p(\theta)$$

The posterior $p(\theta \mid y)$ combines the likelihood with a prior. There is no closed-form expression for the posterior, but you can sample from it.

MARKOV CHAIN MONTE CARLO

Markov chain Monte Carlo builds a chain of parameter vectors $\theta^{(1)}, \theta^{(2)}, \ldots$ whose distribution converges to the posterior.

The simplest version (Metropolis):

1. Propose $\theta' = \theta^{(t)} + \varepsilon$, where ε is a random step
2. Compute the acceptance ratio $\alpha = p(\theta' \mid y) / p(\theta^{(t)} \mid y)$
3. Accept the proposal with probability $\min(1, \alpha)$; otherwise stay at $\theta^{(t)}$

Modern samplers (NUTS) use gradients to propose smarter steps, but the logic is the same: wander through parameter space, spending more time where the posterior is high.

A GEV MODEL IN TURING

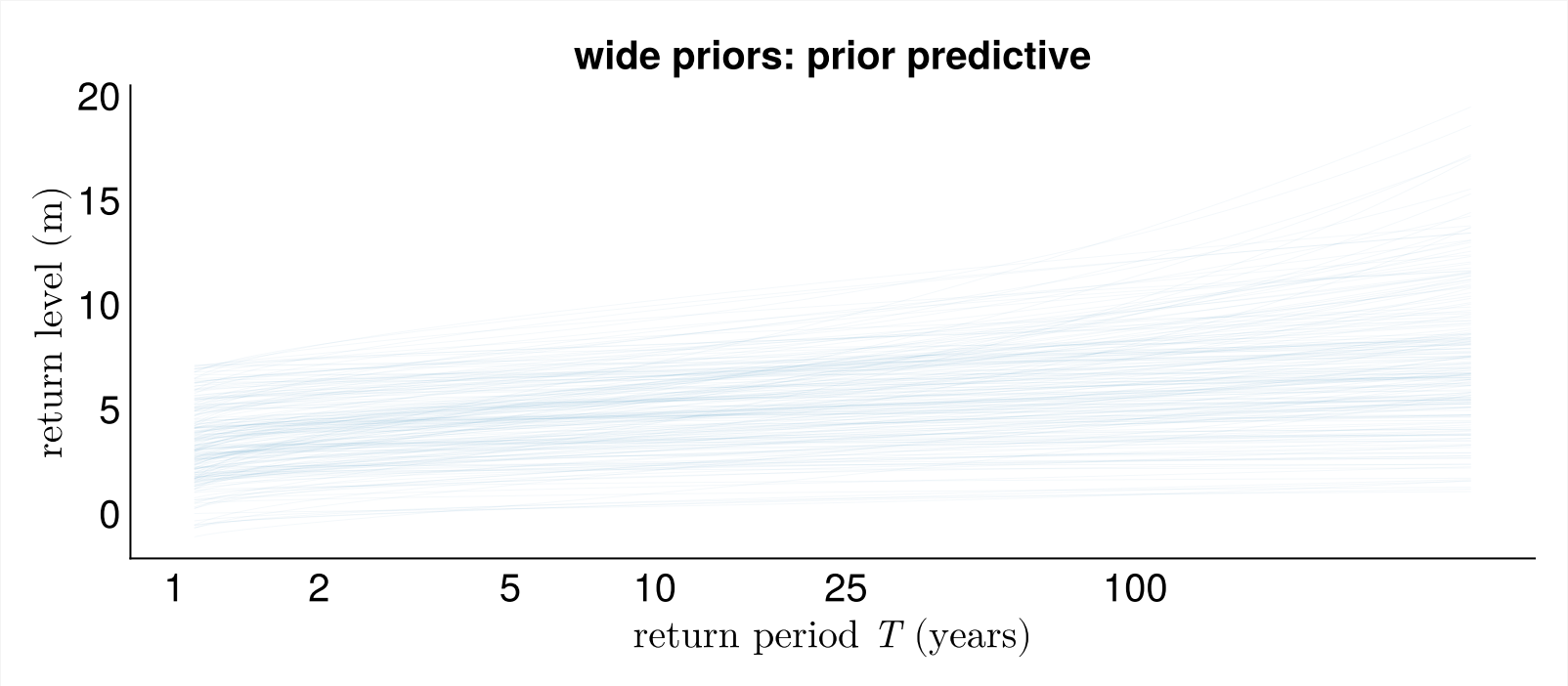
```julia
@model function gev_bayes(y)
    μ ~ Normal(4.0, 2.0)
    σ ~ truncated(Normal(0.2, 0.5); lower=0.001)
    ξ ~ truncated(Normal(0, 0.3); lower=-0.35, upper=0.35)
    if !any(ismissing.(y))
        y .~ GeneralizedExtremeValue(μ, σ, ξ)
    end
end
```

PRIOR PREDICTIVE CHECK: ARE THE WIDE PRIORS REASONABLE?

A prior predictive check asks: do the priors, before seeing any data, produce return-level curves that look like something this site could plausibly produce? Pass `missing` as data so Turing samples everything from the prior.

```
n_miss = length(y)
wide_prior_chain = sample(gev_bayes(fill(missing, n_miss)), Prior(), 1000)
wide_df = DataFrame(wide_prior_chain)
```

PRIOR PREDICTIVE CHECK: ARE THE WIDE PRIORS REASONABLE?



These priors place no constraint on the return levels: the 100-year sea level could be anything from $-10,000$ to $+10,000$ metres.

QUANTILE PRIORS

Instead of reasoning about (μ, σ, ξ) directly, express beliefs about observable quantities.

“The 2-year sea level at Port Pirie is around 4 m.” “The 100-year sea level is probably between 3 and 7 m.”

Translate each into a distribution over the return level at that return period, then add its log-density to the model (Coles, 2001, §9.5; Doss-Gollin & Keller, 2023).

DEFINING QUANTILE PRIORS

```
quantile_priors = [
    (rt=2, dist=Normal(4.0, 1.0)),
    (rt=10, dist=Normal(4.5, 1.5)),
    (rt=100, dist=Normal(5.0, 2.0)),
    (rt=500, dist=Normal(6.0, 3.0)),
]
```

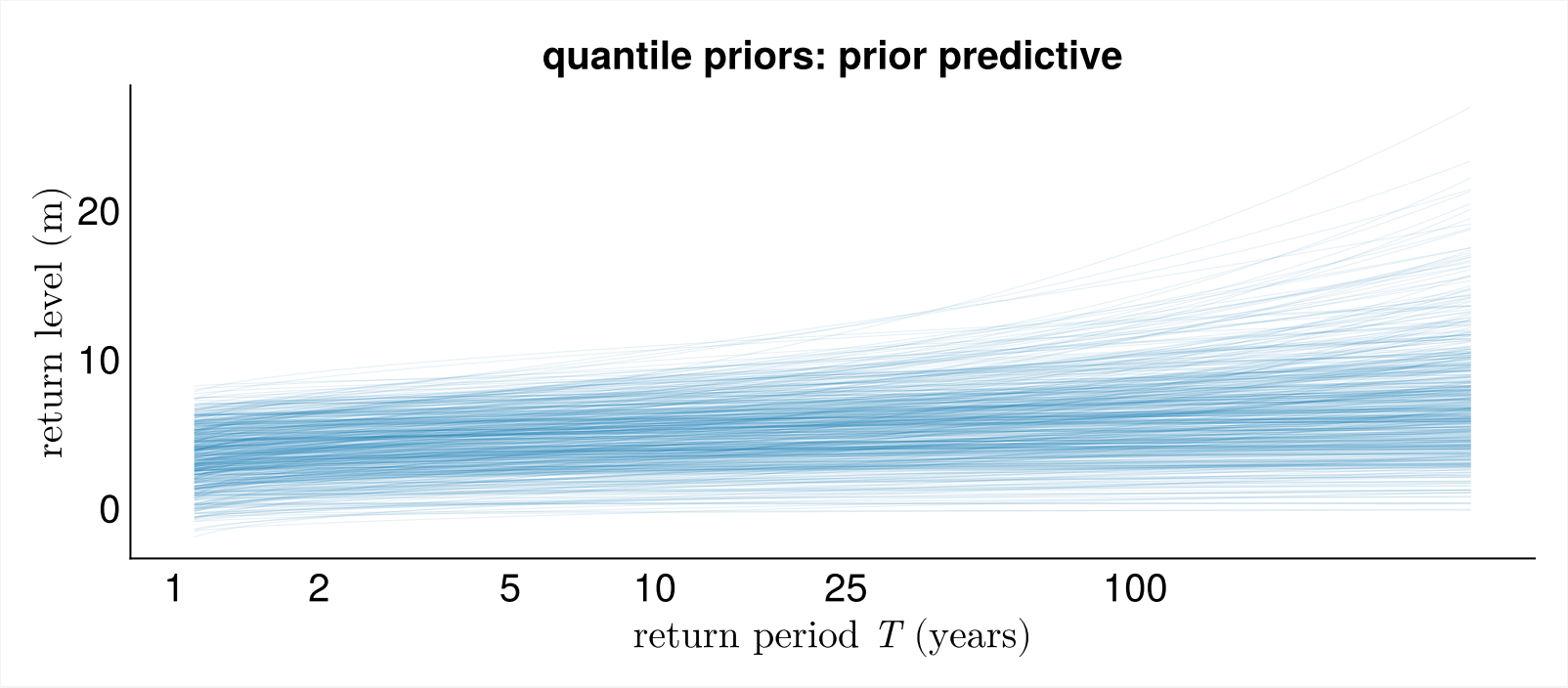
THE MODEL WITH QUANTILE PRIORS

```julia
@model function gev_bayes_qp(y, priors)
    μ ~ Normal(4.0, 2.0)
    σ ~ truncated(Normal(0.2, 0.5); lower=0.001)
    ξ ~ truncated(Normal(0, 0.3); lower=-0.35, upper=0.35)
    dist = GeneralizedExtremeValue(μ, σ, ξ)
    for p in priors
        rl = quantile(dist, 1.0 - 1.0 / p.rt)
        Turing.@addlogprob! logpdf(p.dist, rl)
    end
    if !any(ismissing.(y))
        y .~ dist
    end
end
```

PRIOR PREDICTIVE CHECK: QUANTILE PRIORS

```
1 qp_prior_chain = sample(gev_bayes_qp(fill(missing, n_miss), quantile_priors), Prior(),
2000)
2 qp_df = DataFrame(qp_prior_chain)
```

PRIOR PREDICTIVE CHECK: QUANTILE PRIORS



PRIOR PREDICTIVE CHECK: QUANTILE PRIORS

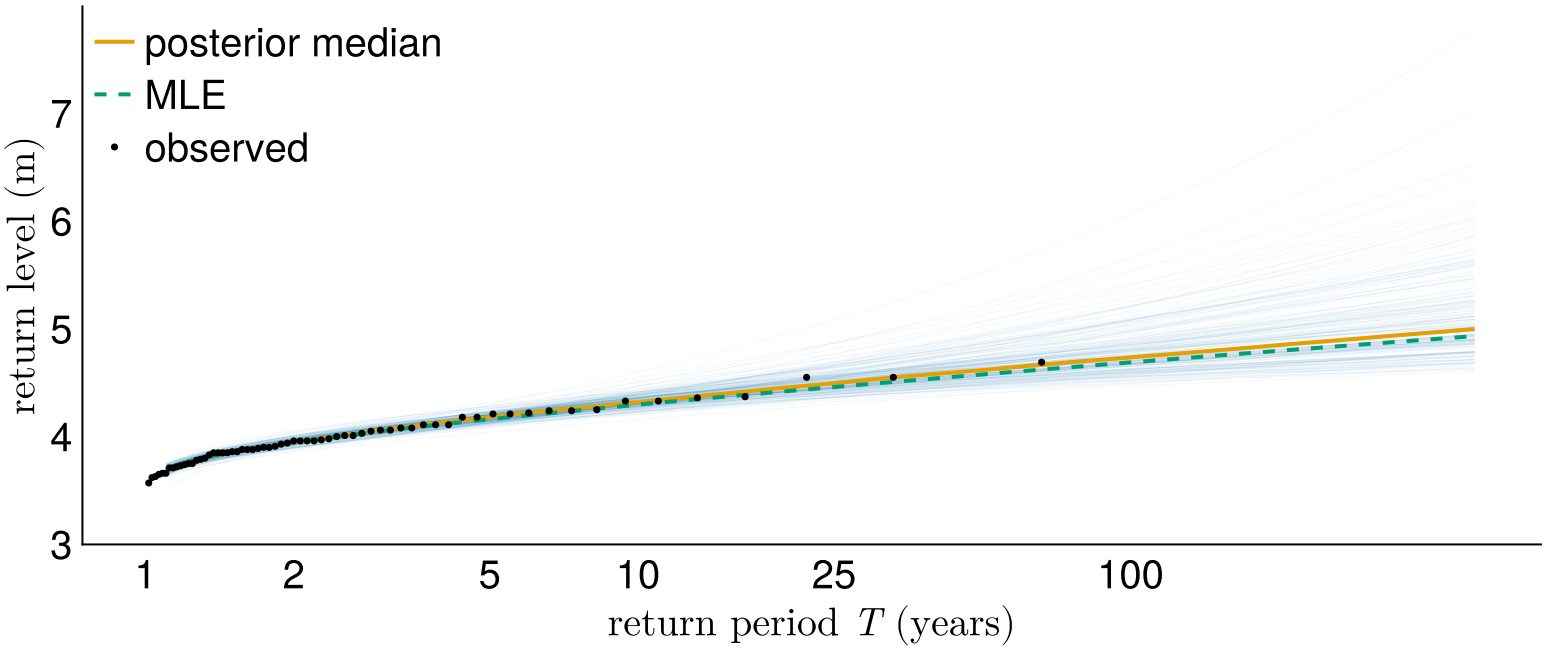
The quantile priors concentrate the curves toward physically plausible sea levels, but some draws still range widely. Compare this to the wide-prior plot: the quantile constraints are doing real work.

POSTERIOR FIT

```
1 chain = sample(gev_bayes_qp(y, quantile_priors), NUTS(0.95), MCMCThreads(), 2_000, 4)
```

```
posterior draws: 8000
```

POSTERIOR RETURN-LEVEL CURVES



MLE VS BAYES: THE 100-YEAR LEVEL

```
MLE:   4.69 m
Bayes: 4.74 m  [4.52, 5.33]  (95% credible)
```

The posterior median is close to the MLE. The credible interval accounts for parameter uncertainty, which the MLE point estimate alone does not.

FRIDAY'S LAB

In the lab you fit a GEV to a NOAA tide gauge four ways:

- `gevfit` and `gevfit_lmom`, as above
- a log-likelihood you write and optimize yourself
- Turing, with wide priors and with quantile priors

Then you change the gauge and see which conclusions survive.